

PRODUCT 03 - Website Care & Uptime Monitoring - Developer Build Sheet

Stack: Frontend Next.js + Backend Node.js + MongoDB

1. Setup

Backend: npm i express mongoose cors dotenv jsonwebtoken bcryptjs axios node-cron pdfkit

Frontend: npx create-next-app@latest frontend --tailwind --app, npm i axios recharts

2. Structure

backend/

app.js, cron.js

models/User.js, Website.js, UptimeLog.js, Incident.js, BusinessApp.js, AuditReport.js

routes/auth.js, websites.js, audit.js, status.js

frontend/

app/login/page.jsx, dashboard/page.jsx, add-website/page.jsx, website/[id]/page.jsx, business-apps/page.jsx, audit/page.jsx,

status-page/[userId]/page.jsx, alerts/settings/page.jsx

lib/api.js

3. Models

User: email, passwordHash

Website: userId, url, name, status up/down, uptimePercent, avgResponseMs, sslExpiryDate, lastCheckedAt

UptimeLog: websiteId, statusCode, responseTimeMs, isUp, checkedAt default now

Incident: websiteId, startedAt, endedAt, durationMinutes, status ongoing/resolved, rootCause

BusinessApp: userId, name, url, type website/tally/erp/cctv/payment_gateway, status, lastChecked

AuditReport: userId, period, uptimePercent, incidents, mttrMinutes, pdfUrl, createdAt

4. Backend APIs

POST /api/auth/register - email, password -> hash -> create User

POST /api/auth/login - email, password -> JWT

POST /api/websites/add - userId from JWT, url, name -> create Website

GET /api/websites/list - userId -> list Websites

GET /api/websites/:id/logs?days=7 - websiteId -> logs last 7 days sorted asc

POST /api/monitor/business-app/add - userId, name, url, type -> create BusinessApp

GET /api/monitor/business-app/list - userId -> list

GET /api/audit/report/generate?userId=.12months - calculate uptime% upLogs/totalLogs, incidents count, MTTR, generate

PDF using pdfkit, save public/reports/, create AuditReport, return pdfUrl

POST /api/alert/whatsapp - websiteId, message -> send WhatsApp via Meta API (for cron)

POST /api/ai/root-cause - statusCode, error -> rules: 0 DNS/Server Down, ssl error SSL Expired, 500 Server Error, timeout

Server Slow -> return rootCause

Cron: node-cron * * * * * every 1 minute -> loop all Website + BusinessApp -> axios.get url timeout 10s -> create UptimeLog
isUp true/false -> update Website status, avgResponseMs, lastCheckedAt -> if down call root-cause -> send WhatsApp alert ->
create Incident with rootCause

5. Frontend Pages

lib/api.js axios baseUrl localhost:5000/api Bearer

login/page.jsx - email, password -> login

dashboard/page.jsx - cards Total Websites, Up, Down, Business Apps -> table Name, URL, Status badge green/red,

Response ms, View link -> fetch /websites/list + /monitor/business-app/list

add-website/page.jsx - Name, URL form -> POST /websites/add -> push dashboard

website/[id]/page.jsx - LineChart recharts responseTimeMs vs checkedAt from /websites/:id/logs, table recent 20 logs Time,
Status, Response

business-apps/page.jsx - Form Name, URL, Type dropdown Website/Tally/ERP/CCTV/Payment -> POST

/monitor/business-app/add, table list

audit/page.jsx - Button Generate ISO Audit Report 12 Months -> GET /audit/report/generate -> show PDF preview + download,
table past reports

status-page/[userId]/page.jsx - Public no auth, fetch websites by userId, show UP/DOWN cards, uptime badge, logo

alerts/settings/page.jsx - Toggles Email, WhatsApp, SMS, Call, Language dropdown, Phone numbers, Test Alert button

6. .env

MONGO_URI=...

JWT_SECRET=...

WHATSAPP_TOKEN=...

7. Run

Backend node app.js starts cron automatically, Frontend npm run dev

8. Done Checklist

- Add website works
- Cron checks every minute, UptimeLog created
- Dashboard shows green/red correctly
- Graph shows response time
- Business app monitoring works same as website
- Audit report PDF generated
- Status page public works